Fixed Effects: Extensions and Inference

SW Chapter 10 (Part 3)

ECON3500: Econometrics and Applications

Spring 2026

Quick Review: What We Covered Thursday

Last class, we covered **difference-in-differences (DiD)**:

Combines a **treatment/control** comparison with a **before/after** comparison

The interaction term $\hat{\delta_1}$ in $y = \beta_0 + \delta_0 \cdot after + \beta_1 \cdot treated + \delta_1 \cdot (after \times treated) + u$ is the DiD estimate

Requires **parallel trends**: in the absence of treatment, both groups would have followed the same trajectory

Pre-trend tests and event study plots provide supporting evidence

Today: extending fixed effects — **time FEs**, **clustered inference**, and where modern DiD is heading.

From Entity FE to Time FE

Why Demeaning Works

Recall from Rae: with **entity fixed effects**, we give each unit its own intercept α_i.

Mathematically, this is **demeaning** — subtracting each entity's time-average from its observations:

$$[y_{it} - \bar{y}_i] = \beta_1 [x_{it} - \bar{x}_i] + [u_{it} - \bar{u}_i]$$

Because $\alpha_i - \bar{\alpha}_i = 0$, the fixed effect drops out completely.

The Within Estimator

This is why FE is called the **within** estimator — we're using variation *within* each entity over time, not variation *between* entities.

Now: What if we also want to control for shocks that hit **all entities equally** in the same period?

Time Fixed Effects

Examples of entity-invariant time shocks:

A national recession hits all cities in 2008

A federal policy change affects all states simultaneously

Inflation affects all firms' costs in the same year

Add **time fixed effects** λ_t: a dummy for each time period.

$$y_{it} = \beta_1 x_{it} + \alpha_i + \lambda_t + u_{it}$$

α_i: entity fixed effect — absorbs entity-specific, time-invariant factors

λ_t: time fixed effect — absorbs time-specific, entity-invariant factors

Connection to DiD

The $\delta_0 \cdot after_t$ term in the DiD regression *is* a (two-period) time fixed effect — it absorbs the common time trend affecting both groups.

What Time FEs Actually Do

A national recession hits **all states** in 2008. Unemployment rises everywhere:

State	2006	2008	Change
Vermont	5.2%	7.8%	+2.6pp
Ohio	5.9%	8.6%	+2.7pp
Texas	4.4%	7.0%	+2.6pp

This common jump of ~2.6pp reflects a **national shock** — not the effect of any state policy. It's **not variation we want** to exploit.

A **year dummy for 2008** absorbs this common shift exactly. After removing the year mean, only **within-year, across-state variation** remains.

The Analogy to Entity FE

Entity FE removes: "Detroit is always high-crime regardless of unemployment" — a unit-specific level

Time FE removes: "2008 was a bad year for everyone" — a year-specific level

Entity + Time Fixed Effects: What's Left?

With **both** entity and time fixed effects:

$$y_{it} = \beta_1 x_{it} + \alpha_i + \lambda_t + u_{it}$$

Controlled for:

All time-invariant entity characteristics (α_i)

All entity-invariant time shocks (λ_t)

Not controlled for:

Factors that vary **across both entities and time**

Knowledge Check

You estimate the effect of a state-level minimum wage increase on employment, with state and year fixed effects. What omitted variable could still bias your results?

. . .

A factor that varies across states *and* over time — e.g., a state-specific economic boom that coincides with the minimum wage increase.

Implementing FE in Stata

Three Ways to Estimate FE

Option 1: Manual Demeaning

Demean your data by hand (subtract entity means), then estimate OLS on the demeaned data.

Option 2: Dummy Variables / `areg`

Include entity dummies directly, or use `areg` to "absorb" one set of fixed effects:

```
areg scrap grant d88 d89, absorb(fcode) cluster(fcode)
```

Option 3: `xtreg` (Recommended)

Declare your panel structure, then estimate with the `fe` option:

```
xtset fcode year
xtreg scrap grant d88 d89, fe vce(cluster fcode)
```

What Entity FE Controls For

Key Principle

Entity fixed effects control for **all time-invariant factors** — observed or unobserved.

You don't need to include time-invariant covariates in an FE model — they are already absorbed.

Practical notes:

We don't usually interpret the fixed effect coefficients themselves

Which entity is the "omitted" group (avoiding the dummy variable trap) doesn't matter

Fixed effects are typically not reported in regression output

The Price of Entity FE

You **cannot** estimate the effect of variables that don't change over time — gender, race, state geography, etc. The fixed effect absorbs them completely.

Inference: Serial Correlation and Clustering

Least Squares Assumptions for Panel Data

$$Y_{it} = \beta_1 X_{it} + \alpha_i + u_{it}, \quad i = 1, \ldots, n; \; t = 1, \ldots, T$$

$E[u_{it} | X_{i1}, \ldots, X_{iT}, \alpha_i] = 0$

- u_{it} cannot be correlated with **any present, past, or future** values of X

$(X_{i1}, \ldots, X_{iT}, u_{i1}, \ldots, u_{iT})$ are i.i.d. draws **across entities**

- Only need independence across entities, not within

(X_{it}, u_{it}) have finite fourth moments

No perfect multicollinearity

If these hold, FE estimates are **consistent and asymptotically normal** for large n.

The Problem of Serial Correlation

$$Y_{it} = \alpha_i + \beta_1 X_{it} + \omega_{it}$$

The time-varying error ω_{it} reflects factors that change over time but affect the outcome.

Example: If Y_{it} is traffic fatalities, maybe ω_{it} includes road quality in state i.

Road quality last year ≈ road quality this year

$\omega_{i,t}$ and $\omega_{i,t-1}$ are **correlated**

This is **autocorrelation** (serial correlation) — observations on the same entity over time are not independent.

Why Autocorrelation Matters

Previously, we assumed observations were independent.

Implausible for data on the **same entity over time**

Without correction, we estimate **incorrect standard errors**

Confidence intervals will **not have 95% coverage**

> **Autocorrelation Does Not Bias $\hat{\beta}$**
>
> The coefficients are still consistent — but our **inference** (standard errors, p-values, confidence intervals) is wrong. Typically: SEs are **too small**, so we reject too often.

The Solution: Clustered Standard Errors

Clustered standard errors allow for arbitrary correlation within entities.

Idea: Allow ω_{it} and ω_{is} (same entity, different times) to be correlated freely, while maintaining independence **across** entities.

In Stata:

```
* With regress:
regress y x i.entity, cluster(entity)

* With areg:
areg y x, absorb(entity) cluster(entity)

* With xtreg (recommended):
xtreg y x, fe vce(cluster entity)
```

⚠ **Cluster in Panel Data**

In this course, if you are using panel data, **cluster standard errors at the entity level** unless there is a strong reason not to. Failing to do so overstates precision and leads to false rejections.

Staggered Adoption and Recent Developments

The Staggered Adoption Problem

The classic DiD has two groups and two periods. In practice, policies are often adopted by **different units at different times** — **staggered adoption**.

Example: States adopt a job training program in different years.

The natural approach: two-way fixed effects (TWFE).

$$y_{it} = \beta_1 \cdot treated_{it} + \alpha_i + \lambda_t + u_{it}$$

Recent research (roughly 2018–2024) has shown that this “obvious” approach can produce **severely misleading estimates** — including **wrong signs**.

A Concrete Example

A job training program with a **true effect that grows over time**: +10 in year 1, +50 in year 2.

	2018	2019	2020
No-treatment baseline	100	105	110
State E (treats in 2019)	100	115	160
State L (treats in 2020)	100	105	120

TWFE takes a **weighted average** of all 2×2 DiD comparisons — including contaminated ones.

Valid comparison (E treated vs. L as control, 2018→2019):

$$\underbrace{(115 - 100)}_{\Delta E} - \underbrace{(105 - 100)}_{\Delta L} = [+10].\,kw$$

Contaminated comparison (L treated vs. already-treated E as "control," 2019→2020):

$$\underbrace{(120 - 105)}_{\Delta L} - \underbrace{(160 - 115)}_{\Delta E} = [-30].alert$$

$$\hat{\beta}_{TWFE} = \frac{1}{2}(+10) + \frac{1}{2}(-30) = [-10].alert$$

The Sign Flipped!

TWFE Gives the Wrong Sign

All true effects are **positive** (+10 or +50), but TWFE estimates **–10**.

Why? State E's outcome rose by 45 from 2019→2020 because its treatment effect *grew*. TWFE treats this as a normal trend, making State L's +15 gain look like a relative decline.

Goodman-Bacon (2021) showed TWFE is a **weighted average** of all possible 2×2 DiD comparisons. When treatment effects **change over time**, already-treated units used as "controls" introduce bias — and can receive **negative weights**.

New Estimators

Several estimators have been developed to handle staggered adoption correctly:

Callaway and Sant'Anna (2021): Group-time-specific effects, then aggregate. Only uses not-yet-treated or never-treated units as controls.

Sun and Abraham (2021): Interaction-weighted estimator correcting for heterogeneous effects across adoption cohorts.

de Chaisemartin and D'Haultfoeuille (2020): Identifies which comparisons get negative weights and proposes robust alternatives.

All share one insight: **never use already-treated units as controls**.

> ⓘ **Beyond ECON3500**
>
> You won't implement these in this course. But if you do applied research with staggered DiD — very common in policy evaluation — you should know about them. Stata packages: `csdid`, `did_multiplegt`, `eventstudyinteract`.

Putting It All Together

Comparison of Panel Data Methods

Method	Data Required	What It Controls	Key Assumption	What Remains
DiD	2 groups, 2 periods	Group-level differences; common time trends	**Parallel trends**	Group-specific time-varying shocks
First Difference	Panel, 2 periods	All time-invariant entity factors	No time-varying OVB	Time-varying omitted variables
Entity FE	Panel, 2+ periods	All time-invariant entity factors	No time-varying OVB	Time-varying omitted variables
Entity + Time FE	Panel, 2+ periods	Time-invariant entity factors + entity-invariant time shocks	No entity-and-time-varying OVB	Entity-time-specific shocks

When to Use What

Difference-in-Differences:

Repeated cross-sections or panel data

Clear treatment/control groups and before/after periods

Requires parallel trends assumption

First Differencing:

Panel data with 2 periods

Equivalent to entity FE with 2 periods; simple and transparent

Fixed Effects:

Panel data with 2+ periods

Include entity FE, time FE, or both depending on threats

More efficient than first differencing when $T > 2$ and errors are not serially correlated

Common Pitfalls

⚠ 1. Forgetting to Cluster

Panel data almost always requires clustered standard errors. Unclustered SEs will be **too small** — false precision.

⚠ 2. Trying to Estimate Time-Invariant Effects

With entity FE, you **cannot** estimate the effect of variables that don't vary over time. The fixed effect absorbs them.

⚠ 3. Assuming FE Solves All OVB

Fixed effects only eliminate **time-invariant** omitted variables. Time-varying confounders can still bias your estimates.

Key Takeaways

Time fixed effects λ_t absorb entity-invariant time shocks — the same demeaning logic applied across time

With entity + time FEs, only **entity-and-time-varying** confounders remain a threat

Panel data creates **serial correlation** — always use **clustered standard errors** at the entity level

FE **cannot estimate** effects of time-invariant variables — the price of eliminating time-invariant OVB

With **staggered adoption**, standard TWFE can give misleading results — modern estimators (Callaway-Sant'Anna, Sun-Abraham) fix this by never using already-treated units as controls